



VIT
UNIVERSITY
(Estd. u/s 3 of UGC Act 1956)



Vellore - 632 014, Tamil Nadu, India

SCHOOL OF COMPUTING SCIENCE AND ENGINEERING

DIGITAL ASSIGNMENT- I , WINTER 2022-23

Programme : B.Tech

Max.Marks : 10

Course : Data Structures and Algorithms

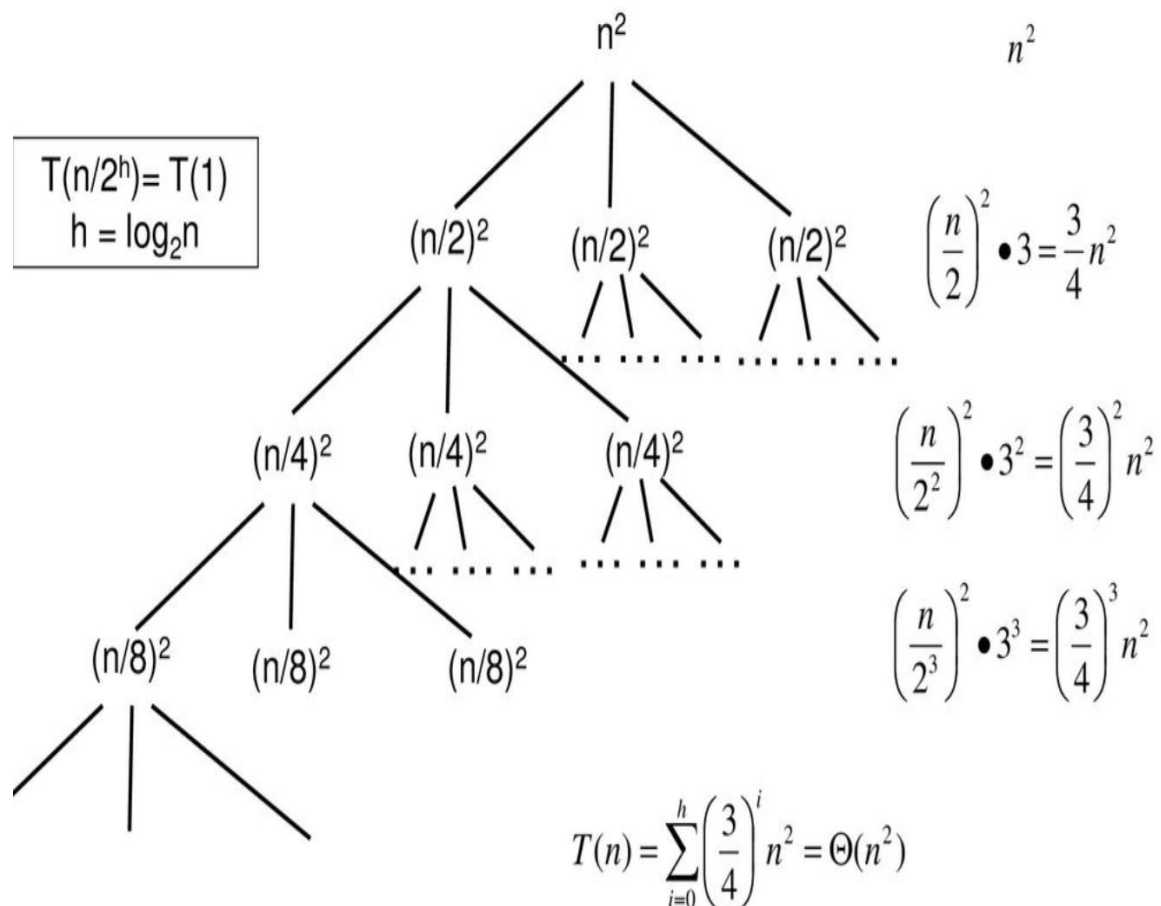
Course Code : CSE202L

Questions

1. Find the complexity of the below recurrence equation using the tree method

$$T(n) = 3T(n/2) + n^2$$

Solution:



2. Find the complexity of the below recurrence equation using the tree method

$$T(n) = T(n/4) + T(n/2) + n^2$$

Solution:

$$\begin{array}{rcl}
 & n^2 & \dots n^2 \\
 & \swarrow \quad \searrow & \\
 (n/4)^2 & & (n/2)^2 \dots \frac{5}{16}n^2 \\
 \swarrow \quad \searrow & & \swarrow \quad \searrow \\
 (n/16)^2 & (n/8)^2 & (n/8)^2 & (n/4)^2 \dots \frac{25}{256}n^2 \\
 \vdots & & & \vdots \\
 / & & & \\
 \Theta(1) & & &
 \end{array}$$

Total = $n^2 \left(1 + \frac{5}{16} + \left(\frac{5}{16} \right)^2 + \left(\frac{5}{16} \right)^3 + \dots \right)$
 $= \Theta(n^2)$ *geometric series*

$$1 + x + x^2 + \dots + x^n = \frac{1 - x^{n+1}}{1 - x} \text{ for } x \neq 1$$

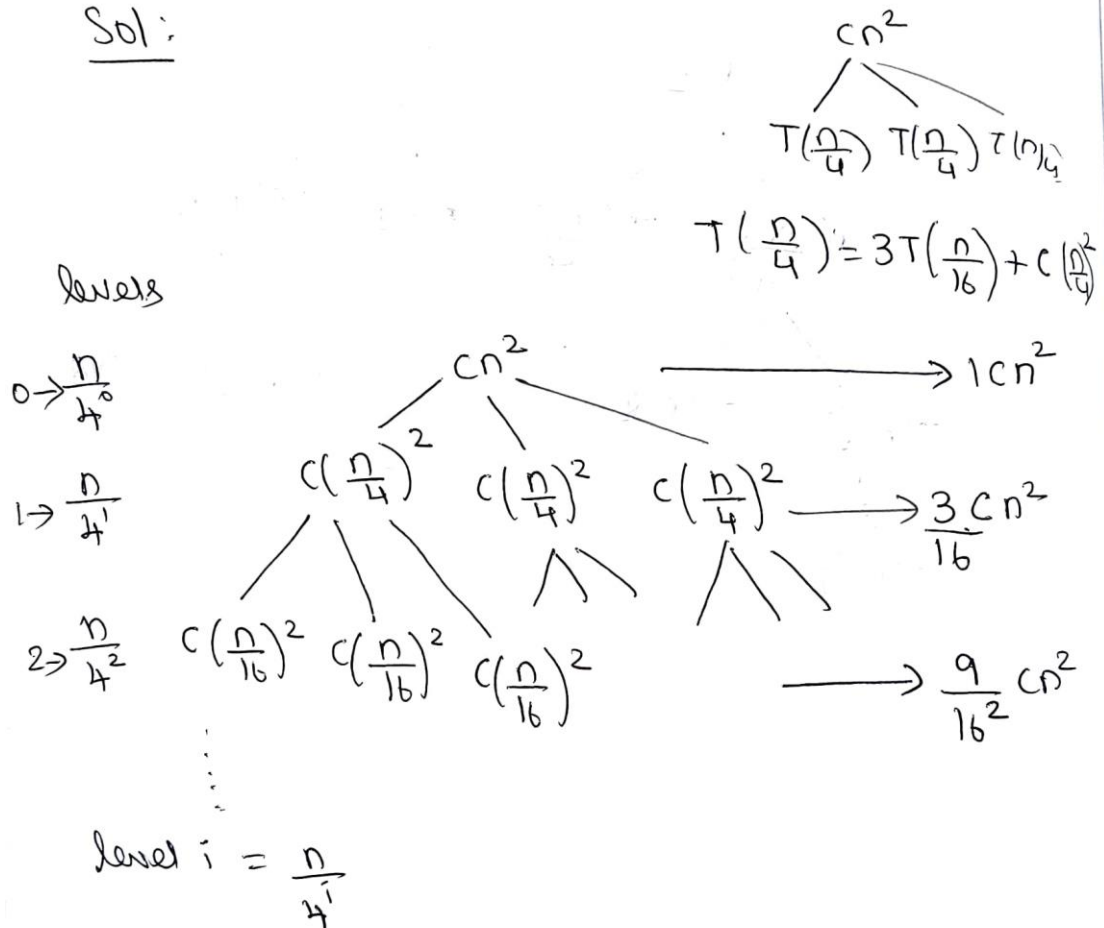
$$1 + x + x^2 + \dots = \frac{1}{1 - x} \text{ for } |x| < 1$$

3. Find the complexity of the below recurrence equation using the tree method

$$T(n) = 3T(n/4) + cn^2$$

Solution:

Sol:



$$\frac{n}{4^i} = 1$$

$$n = 4^i$$

Taking log on both sides

$$i = \log_4 n$$

$$\log n = \log 4^i$$

$$\log n = i \log 4$$

$$i = \log_4 n$$

Here,

$$\text{level } 0 = 3^0 = 1$$

$$\text{level } 1 = 3^1 = 3$$

$$\text{level } 2 = 3^2 = 9$$

$\vdots$

$$\begin{aligned}\text{level } i &= 3^i = 3^{\log_4 n} \\ &= n^{\log_4 3}\end{aligned}$$

$$\begin{aligned}\text{Cost of last level} &= n^{\log_4 3} \times T(1) \\ &= \Theta(n^{\log_4 3})\end{aligned}$$

Add cost of all levels

$$= \left[cn^2 + \frac{3}{16} cn^2 + \frac{9}{16^2} cn^2 + \dots \right] + \Theta(n^{\log_4 3})$$

$$= cn^2 \left[1 + \frac{3}{16} + \frac{9}{16^2} + \dots \right] + \Theta(n^{\log_4 3})$$

$$\underbrace{\hspace{10em}}_{a, ar, ar^2, \dots}$$

$$S_n = \frac{a}{1-r} - \frac{ar^n}{1-r}$$

$$1 + \frac{3}{16} + \frac{9}{16^2} + \dots$$

$$= \frac{a}{1-r} - \frac{ar^n}{1-r} \quad r = \frac{a_2}{a_1} = \frac{\frac{3}{16}}{1} = \frac{3}{16}$$

$$= \frac{1}{1 - \frac{3}{16}} - \frac{1 \cdot \left(\frac{3}{16}\right)^i}{1 - \frac{3}{16}} \quad i \rightarrow i \text{ level}$$

$$= \frac{1}{\frac{16-3}{16}} - \frac{\left(\frac{3}{16}\right)^{\log_4 n}}{\frac{16-3}{16}}$$

$$= \frac{16}{13} - \frac{16}{13} \left(\frac{3}{16}\right)^{\log_4 n}$$

$$= \frac{16}{13} \left(1 - \left(\frac{3}{16}\right)^{\log_4 n}\right)$$

$$cn^2 \left[1 + \frac{3}{16} + \frac{9}{16^2} + \dots\right] + \Theta(n^{\log_4 3})$$

$$\frac{16}{13} cn^2 \left(1 - \left(\frac{13}{16}\right)^{\log_4 n}\right) + \Theta(n^{\log_4 3})$$

$$\frac{16}{13} cn^2 - \frac{16}{13} cn^2 \cdot \left(\frac{13}{16}\right)^{\log_4 n} + \Theta(n^{\log_4 3})$$

$$= \Theta(n^2)$$

4. Find the complexity of the below recurrence equation using the back substitution method

$$T(n) = T(n/2) + c \quad \text{if } n > 1$$

$$T(1) = 1 \quad \text{if } n = 1$$

Solution:

$$T(n) = T(n/2) + c \longrightarrow \textcircled{1}$$

$$T\left(\frac{n}{2}\right) = T\left(\frac{n}{2} \times \frac{1}{2}\right) + c$$

$$= T\left(\frac{n}{4}\right) + c \longrightarrow \textcircled{2}$$

$$T\left(\frac{n}{4}\right) = T\left(\frac{n}{8}\right) + c \longrightarrow \textcircled{3}$$

Backward
Substitution.

Substitute $\textcircled{2}$ in $\textcircled{1}$

$$T(n) = T\left(\frac{n}{4}\right) + c + c$$

$$= T\left(\frac{n}{2^2}\right) + 2c$$

$$= T\left(\frac{n}{8}\right) + 3c$$

$$= T\left(\frac{n}{2^3}\right) + 3c$$

$$= T\left(\frac{n}{2^4}\right) + 4c$$

$$\vdots$$

K steps $T\left(\frac{n}{2^K}\right) + Kc$

Now, we should use the boundary (base) condition,
(i.e) $T(1) = 1$

In order to use the boundary condition, the
entity inside $T()$ must be 1.

$$T\left(\frac{n}{2^k}\right) + kc \longrightarrow \textcircled{4}$$

$\longrightarrow$ To make $T(1)$

We assume,

$$n = 2^k \text{ Substitute in } \textcircled{4}$$

$$T\left(\frac{n}{n}\right) + kc$$

$$T(1) + kc$$

$$1 + kc$$

Now, calculate the value of 'k', take log on
both sides

$$n = 2^k$$

$$k = \log_2 n$$

$$O(\log n)$$

Logarithmic form	$\log_2 N = k$
Exponential form	$2^k = N$

5. Find the complexity of the below recurrence equation using the master method

$$T(n) = T(n) = 2T(n/2) + \sqrt{n}$$

Solution: $T(n) = 2T(n/2) + \sqrt{n}$, the values of $a = 2$, $b = 2$ and $k = 1/2$.

Here $\log_b(a) = \log_2(2) = 1 > k$.

Therefore, the complexity will be $\Theta(n)$